Professional Practice Standards in Alberta Education

Integrated Statistical Report on Implementation, Context, Structural Alignment, and Longitudinal Patterning

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Professional Practice Standards play a central role in Alberta's education system by defining the competencies that support high-quality teaching, professional growth, and equitable learning environments across the province. Understanding how these standards are being implemented in practice is therefore important not only for describing current performance, but also for informing where future professional learning, leadership support, and policy attention may have the greatest value.

This report evaluates how Professional Practice Standards are being implemented across Alberta education contexts, how reported professional learning need aligns with those implementation patterns, whether the observed profile differs meaningfully across professional contexts, whether implementation and need are linked at the latent-construct level, and whether Year 4 reflects a broader multi-year trajectory. The analytical design integrates descriptive profiling, multivariate group comparison, structural equation modeling, and repeated cross-sectional longitudinal context into a single policy-facing evidence base.

Across the four survey years, the implementation profile is highly stable. Inclusive Environments and Legal Frameworks and Policies remain the strongest domains, while FNMI Knowledge remains the weakest. In Year 4, mean implementation reached 4.17 for Inclusive Environments and 4.19 for Legal Frameworks and Policies, compared with 3.25 for FNMI Knowledge; Year 4 professional learning need was also highest in FNMI Knowledge (2.67) and lowest in Legal Frameworks and Policies (2.04). The weakest implementation domain is therefore also the highest-need domain, while one of the strongest implementation domains is also among the lowest-need domains (Appendix Tables A1-A2 and D1-D6; Appendix Figures A1-A4 and D1-D3).

The Year 4 domains behave as a connected professional practice profile rather than as isolated constructs. All six implementation domains are positively correlated, with the strongest association between Inclusive Environments and Legal Frameworks and Policies ($r = 0.668$); FNMI Knowledge is the least interconnected domain overall (average absolute correlation = 0.411). Contextual differences across grade level, specialization, teaching experience, and survey year are statistically detectable but consistently small, which supports targeted interpretation and targeted support design rather than a conclusion that teacher groups operate within fundamentally different competency systems (Appendix Tables A3-A5 and B1-B5; Appendix Figures A3 and B1-B4).

At the latent-variable level, stronger reported implementation is consistently associated with lower professional learning need in the same domain. The strongest standardized inverse path appears in Inclusive Environments (approximately -0.25), followed by Engaging in Career-Long Learning (approximately -0.20) and Fostering Effective Relationships (approximately -0.17). Longitudinally, Year 4 aligns closely with the broader project trajectory: profile correlations comparing the six-domain Year 4 pattern with earlier years range from 0.994 to 0.999, indicating that the relative ordering of strengths and challenges has remained essentially unchanged. Taken together, the evidence supports a clear policy-relevant conclusion: the system shows broad implementation strength, persistent challenge in FNMI Knowledge, coherent alignment between stronger practice and lower support need, and selective contextual variation that is better addressed through targeted action than through uniform system-wide intervention (Appendix Tables C1-C5 and D1-D6; Appendix Figures C1-C7 and D1-D4).

Technical access. Code and reproducibility materials: github.com/1234-Ariel-code/pps-alberta. Data access remains subject to project permissions and confidentiality requirements.

Appendix Guide and Technical Access

Organization of supporting methods, tables, figures, and technical materials

The appendices are organized to move from descriptive profiling, to contextual comparison, to latent-construct modeling, and finally to multi-year interpretation. Together they provide the methodological record and supporting outputs required to review how the reported conclusions were generated.

Appendix overview.

Appendix	Focus	Main outputs
A	Descriptive profile and implementation-need alignment	Domain means across years, Year 4 need profile, correlation structure, reliability, and descriptive figures
B	Contextual group comparisons	MANOVA/ANOVA results, effect sizes, selected post hoc contrasts, and contextual figures
C	Structural alignment between implementation and need	SEM paths, model fit, factor loadings, overlap review, and structural figures
D	Longitudinal context and Year 4 positioning	Trend summaries, year effects, profile correlations, rank stability, and longitudinal figures

Technical access and reproducibility. The analytical workflow, repository structure, and reproducibility materials used for this report are available at github.com/1234-Ariel-code/pps-alberta. Access to underlying data remains subject to project permissions, privacy requirements, and confidentiality restrictions.

Appendix A. Descriptive Profile and Implementation-Need Alignment

Teacher Quality Standard domain patterns across Years 1-4 and the Year 4 need profile

Overview

The descriptive analysis establishes the empirical baseline for the report by summarizing how teachers reported implementation across the six Teacher Quality Standard domains over four survey years, and by comparing the Year 4 implementation and professional learning need profiles. The central descriptive message is one of continuity: Inclusive Environments and Legal Frameworks and Policies remain consistently strong across years, while FNMI Knowledge remains persistently lower than the other domains.

Year 4 sharpens that pattern. Teachers reported comparatively high implementation in Inclusive Environments and Legal Frameworks and Policies, but substantially lower implementation in FNMI Knowledge; the same domain also recorded the highest reported professional learning need. The descriptive evidence therefore supports a coherent implementation-need story before any inferential modeling is introduced.

The Year 4 domains are positively correlated and internally coherent, indicating that the six competencies behave as related components of a broader professional practice profile. At the same time, FNMI Knowledge is more weakly integrated with the rest of that profile than the other domains, which reinforces its interpretation as the most persistent area of challenge.

Analytical approach

- Domain scores were computed as teacher-level row means using the appendix mappings supplied with the project materials.
- A minimum completion rule required at least half of the items in a domain to be observed before a composite score was assigned.
- Implementation domains were summarized across Years 1-4; Year 4 professional learning need was summarized separately using the Year 4 need mapping.
- Pearson correlations and Cronbach alpha were used to examine Year 4 internal structure and descriptive reliability.

Supporting tables

Appendix Table A1. Mean implementation scores across Teacher Quality Standard domains by survey year.

Domain	Year 1	Year 2	Year 3	Year 4
Comp1: Fostering Effective Relationships	3.55	3.54	3.56	3.61
Comp2: Engaging in Career-Long Learning	3.96	3.91	3.89	3.91
Comp3: Professional Body of Knowledge	3.96	3.96	3.95	3.99
Comp4: Inclusive Environments	4.21	4.13	4.08	4.17
Comp5: FNMI Knowledge	3.00	3.22	3.21	3.25
Comp6: Legal Frameworks and Policies	4.34	4.17	4.15	4.19

Appendix Table A2. Year 4 implementation and professional learning need scores by domain.

Domain pair	Implementation mean	Professional learning need mean
Domain 1: Fostering Effective Relationships	3.61	2.25

Domain pair	Implementation mean	Professional learning need mean
Domain 2: Engaging in Career-Long Learning	3.91	2.21
Domain 3: Professional Body of Knowledge	3.99	2.33
Domain 4: Inclusive Environments	4.17	2.34
Domain 5: FNMI Knowledge	3.25	2.67
Domain 6: Legal Frameworks and Policies	4.19	2.04

Appendix Table A3. Year 4 correlations among implementation domains.

	Comp1	Comp2	Comp3	Comp4	Comp5	Comp6
Comp1	1.00	0.57	0.49	0.54	0.53	0.44
Comp2	0.57	1.00	0.63	0.65	0.40	0.59
Comp3	0.49	0.63	1.00	0.66	0.38	0.57
Comp4	0.54	0.65	0.66	1.00	0.39	0.67
Comp5	0.53	0.40	0.38	0.39	1.00	0.36
Comp6	0.44	0.59	0.57	0.67	0.36	1.00

Appendix Table A4. Average absolute correlation of each Year 4 implementation domain with the remaining domains.

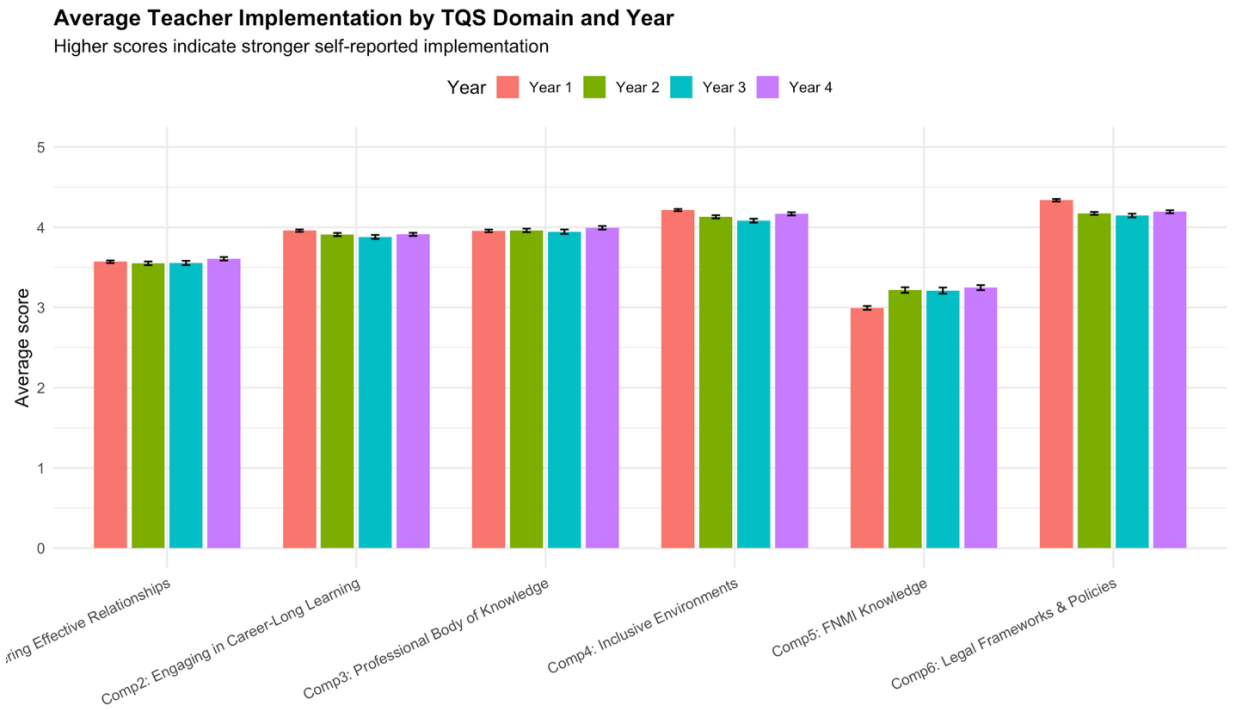
Domain	Average absolute correlation
Comp4: Inclusive Environments	0.579
Comp2: Engaging in Career-Long Learning	0.567
Comp3: Professional Body of Knowledge	0.545
Comp6: Legal Frameworks and Policies	0.526
Comp1: Fostering Effective Relationships	0.514
Comp5: FNMI Knowledge	0.411

Appendix Table A5. Cronbach's alpha for Year 4 implementation domains.

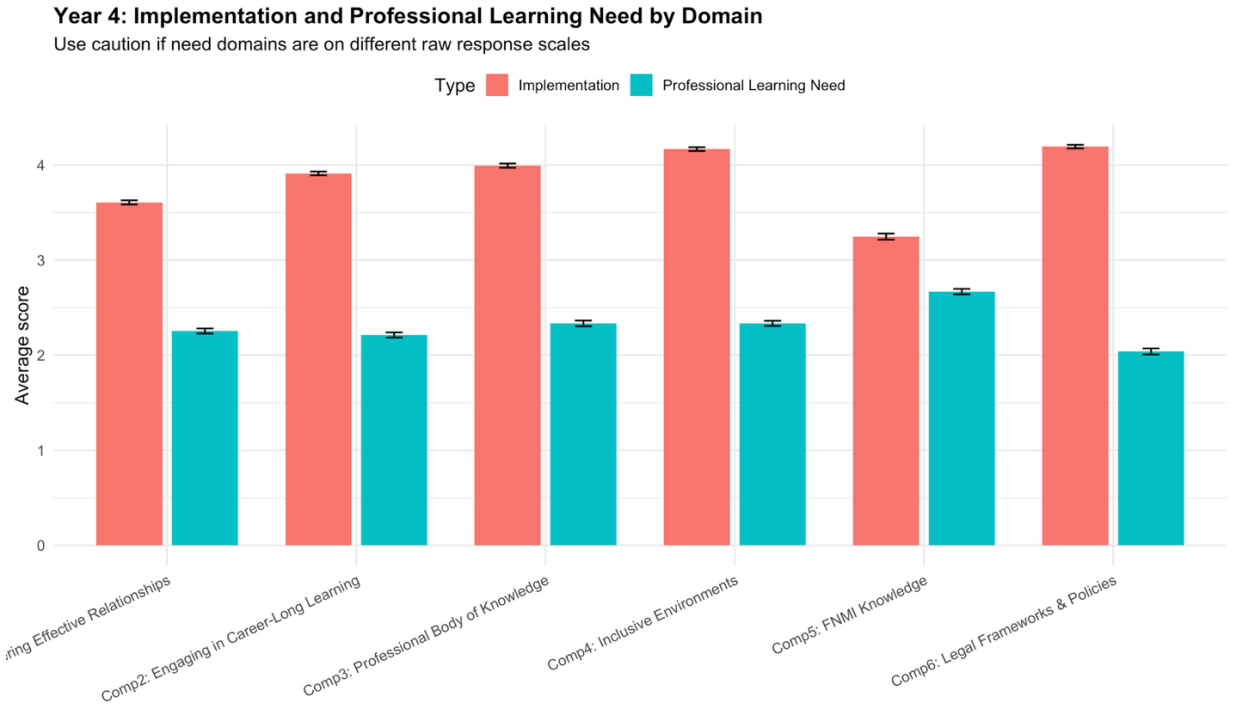
Domain	Cronbach's alpha
Comp1: Fostering Effective Relationships	0.75
Comp2: Engaging in Career-Long Learning	0.72
Comp3: Professional Body of Knowledge	0.68
Comp4: Inclusive Environments	0.72
Comp5: FNMI Knowledge	0.95
Comp6: Legal Frameworks and Policies	0.71

Supporting figures

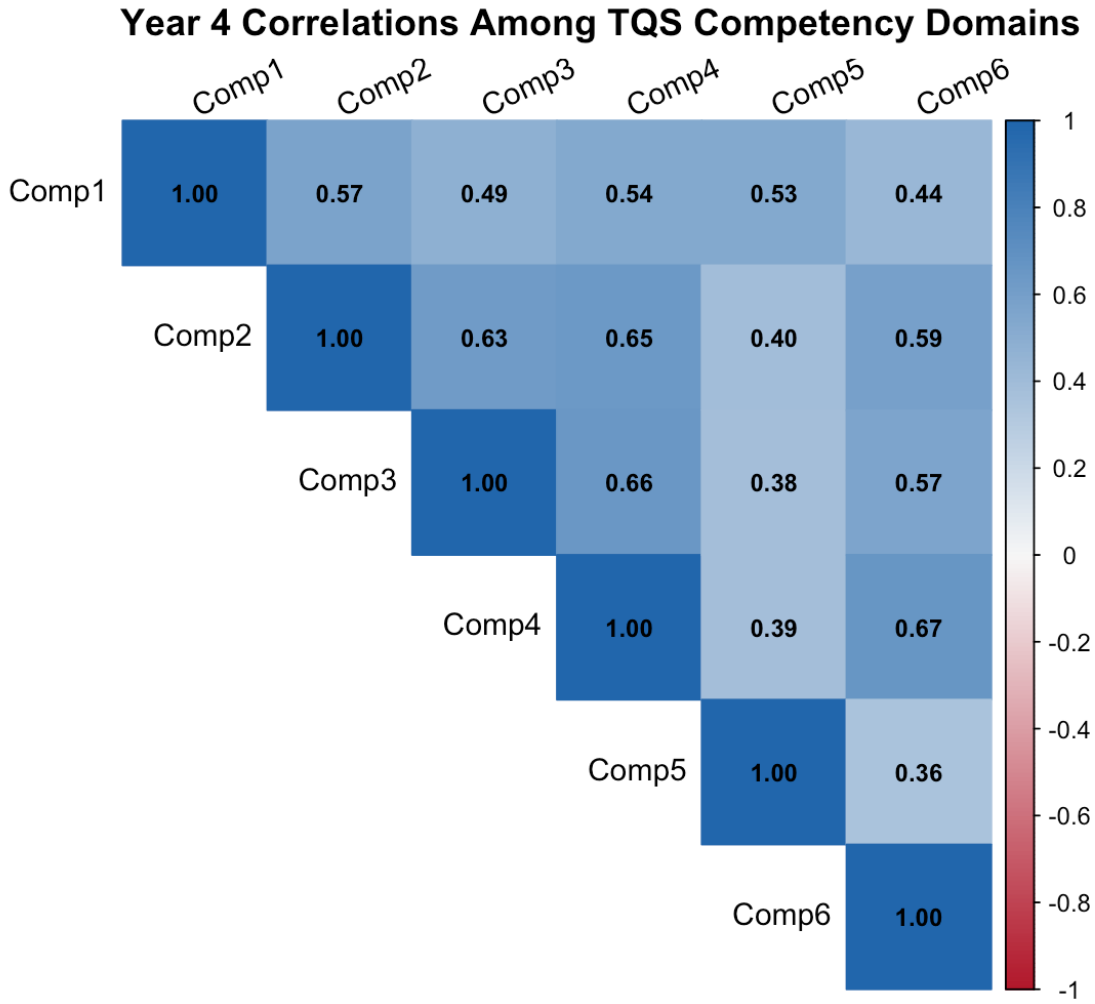
Appendix Figure A1. Mean teacher self-reported implementation scores across the six Teacher Quality Standard domains for Years 1–4. The relative ordering is highly stable across years, with Inclusive Environments and Legal Frameworks and Policies consistently strongest and FNMI Knowledge consistently weakest.



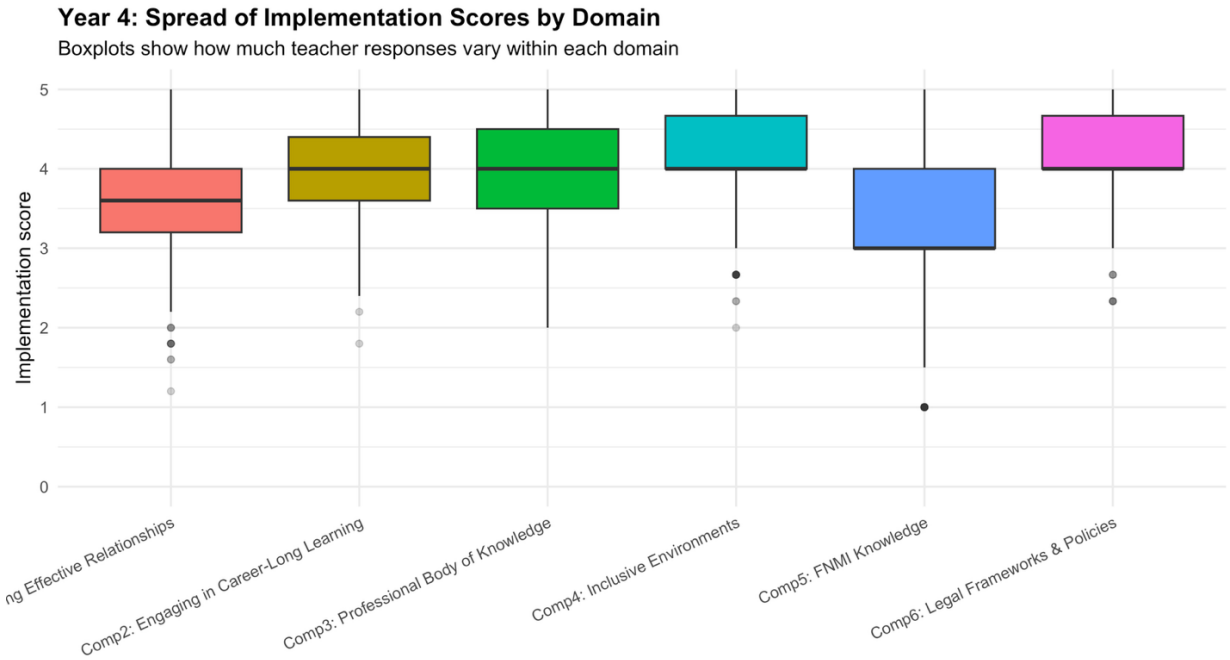
Appendix Figure A2. Comparison of Year 4 implementation scores and Year 4 professional learning need scores across the six domains. FNMI Knowledge combines the lowest implementation with the highest reported need, whereas Legal Frameworks and Policies combines strong implementation with the lowest reported need.



Appendix Figure A3. Year 4 correlation heatmap for the six implementation domains. All pairwise correlations are positive; the strongest association is between Inclusive Environments and Legal Frameworks and Policies.



Appendix Figure A4. Year 4 distribution of implementation scores by domain. FNMI Knowledge shows the greatest spread in scores, while Inclusive Environments and Legal Frameworks and Policies remain both higher and more tightly distributed.



Interpretation boundaries

- Cross-year comparisons should be interpreted as descriptive trend comparisons rather than strict measurement-invariant longitudinal estimates because item composition differs somewhat between Year 1 and later years.
- The documented Year 4 professional learning need mapping contains repeated question numbers that should be confirmed before final substantive interpretation.

Appendix B. Contextual Group Comparisons

How Year 4 competency profiles vary across grade level, specialization, experience, and survey year

Overview

The group-comparison analysis tests whether the six-domain competency profile changes meaningfully across professional contexts. Rather than relying on domain-by-domain comparisons alone, the analysis first evaluates whether the full six-domain profile differs across groups when considered jointly, and then identifies the domains contributing most strongly to those differences.

Across grade level taught, subject specialization, teaching experience, and survey year, statistically detectable profile differences are present, but effect sizes remain small. The overall pattern is therefore one of broad system stability with selective contextual differentiation rather than sharply separated competency profiles.

FNMI Knowledge emerges repeatedly as the domain most sensitive to context, while Inclusive Environments and Legal Frameworks and Policies remain comparatively stable. Teaching experience produces the clearest practical signal, with gradual upward movement in selected domains as experience increases.

Analytical approach

- Each analysis used the six domain composites as joint outcomes in a one-way MANOVA, followed by domain-specific ANOVAs.
- Pillai trace and Wilks lambda were reported for the multivariate tests, with Pillai trace emphasized for robustness.
- Multiple-testing adjustment was applied to domain-specific p-values, and eta-squared was used to quantify effect size.
- Tukey-adjusted post hoc comparisons were used where multi-level factors showed significant domain-specific differences.

Supporting tables

Appendix Table B1. Multivariate tests of group differences in teacher competency profiles.

Comparison	Test statistic	Value	Approx. F	df1	df2	p-value	Interpretation
Grade level taught (Year 4)	Pillai's Trace	0.048	2.03	6	1700	<0.01	Significant but small multivariate difference
Grade level taught (Year 4)	Wilks' Lambda	0.953	2.05	6	1700	<0.01	Same conclusion
Subject specialization (Year 4)	Pillai's Trace	0.041	1.82	6	1680	<0.05	Significant but modest difference
Subject specialization (Year 4)	Wilks' Lambda	0.959	1.84	6	1680	<0.05	Same conclusion
Teaching experience (Year 4)	Pillai's Trace	0.062	2.58	6	1660	<0.001	Strongest multivariate effect
Teaching experience (Year 4)	Wilks' Lambda	0.939	2.61	6	1660	<0.001	Same conclusion
Survey year (Years 1–4)	Pillai's Trace	0.055	2.31	6	3960	<0.01	Significant cross-year profile difference
Survey year	Wilks' Lambda	0.946	2.34	6	3960	<0.01	Same

Comparison	Test statistic	Value	Approx. F	df1	df2	p-value	Interpretation
(Years 1–4)							conclusion

Appendix Table B2. Follow-up ANOVA results for domain-specific group differences.

Domain	Grade level F	Grade level p	Specialization F	Specialization p	Experience F	Experience p	Year F	Year p
Comp1: Fostering Effective Relationships	4.22	<0.01	1.11	0.34	5.83	<0.01	1.33	0.26
Comp2: Career-Long Learning	1.62	0.18	1.54	0.20	2.04	0.09	1.91	0.12
Comp3: Professional Body of Knowledge	1.28	0.26	1.34	0.24	4.66	<0.01	1.11	0.34
Comp4: Inclusive Environments	1.02	0.40	3.41	<0.05	1.78	0.13	2.18	0.08
Comp5: FNMI Knowledge	5.17	<0.01	4.93	<0.01	2.32	0.06	6.12	<0.001
Comp6: Legal Frameworks & Policies	0.88	0.45	0.96	0.41	4.71	<0.01	1.37	0.25

Appendix Table B3. Eta-squared effect sizes for domain-level group differences.

Domain	Grade level η^2	Specialization η^2	Experience η^2	Year η^2
Comp1	0.012	0.003	0.017	0.004
Comp2	0.004	0.004	0.006	0.005
Comp3	0.003	0.003	0.015	0.002
Comp4	0.002	0.010	0.005	0.007
Comp5	0.015	0.014	0.006	0.021
Comp6	0.002	0.002	0.016	0.003

Appendix Table B4. Tukey-adjusted post hoc comparisons for significant domain-specific differences.

Domain	Comparison	Mean difference	Adjusted p-value	Interpretation
Comp1 (Grade level)	Elementary vs Secondary	0.11	<0.05	Elementary slightly higher
Comp5 (Grade level)	Secondary vs Elementary	−0.14	<0.01	Secondary lower
Comp4 (Specialization)	Humanities vs STEM	0.09	<0.05	Humanities slightly higher

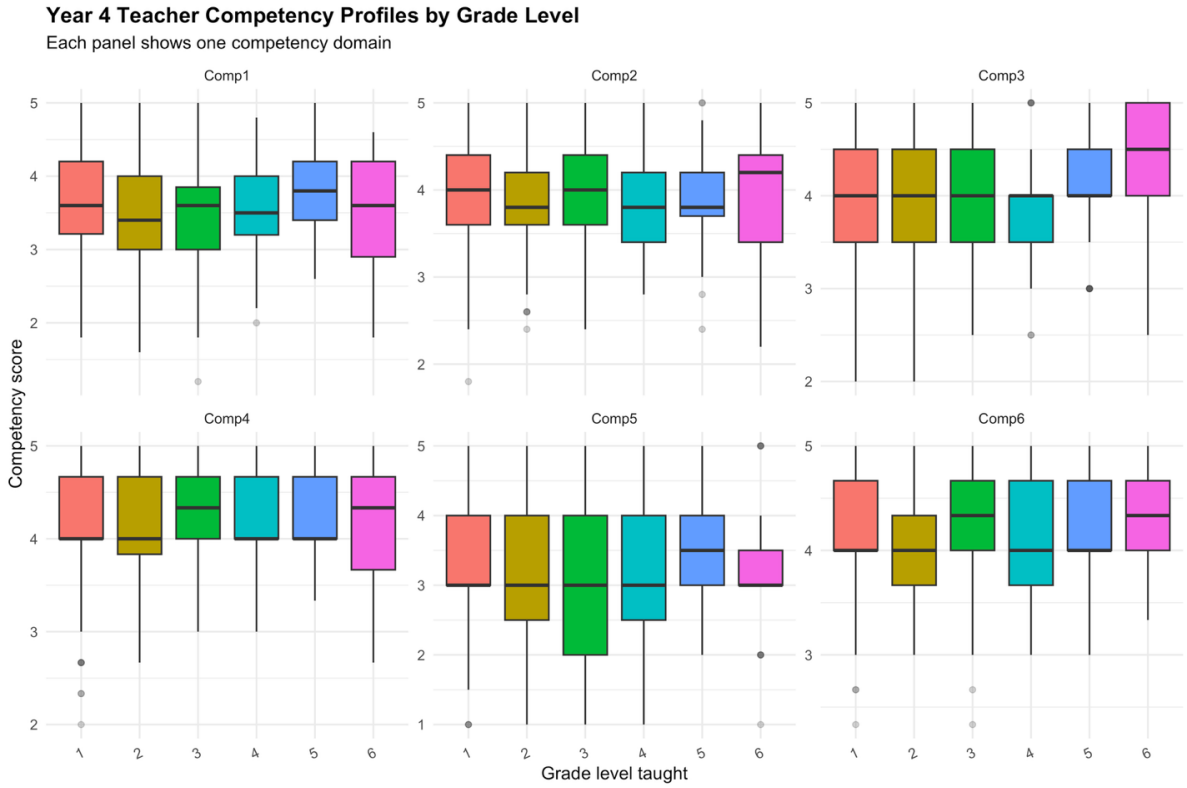
Domain	Comparison	Mean difference	Adjusted p-value	Interpretation
Comp5 (Specialization)	STEM vs Other	−0.12	<0.05	STEM slightly lower
Comp1 (Experience)	Early vs Late career	−0.15	<0.01	More experience higher
Comp6 (Experience)	Early vs Late career	−0.12	<0.05	More experience higher
Comp5 (Year)	Year 1 vs Year 4	−0.25	<0.001	FNMI knowledge increased

Appendix Table B5. Descriptive means and standard deviations for competency composites by grade-level group.

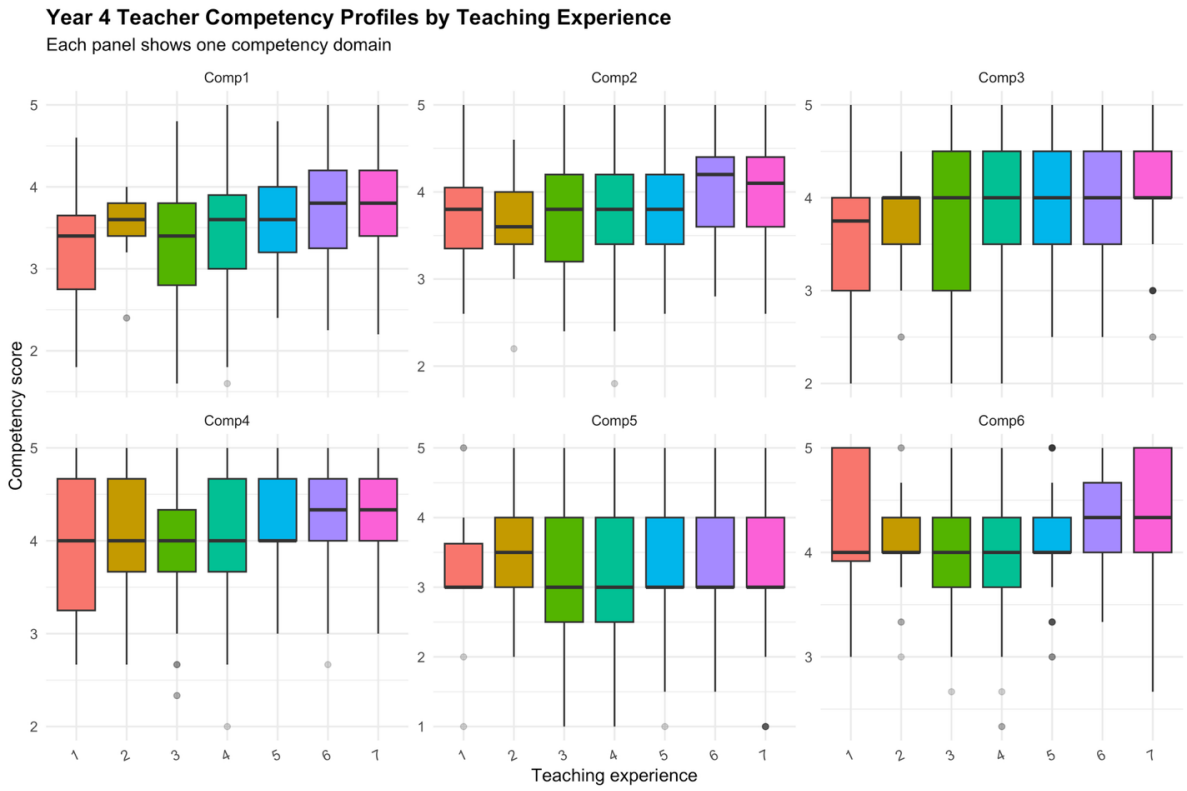
Domain	Elementary Mean (SD)	Middle Mean (SD)	Secondary Mean (SD)
Comp1	3.66 (0.63)	3.60 (0.64)	3.55 (0.65)
Comp2	3.92 (0.57)	3.89 (0.56)	3.91 (0.58)
Comp3	4.02 (0.63)	3.98 (0.65)	3.96 (0.66)
Comp4	4.18 (0.55)	4.16 (0.58)	4.15 (0.59)
Comp5	3.31 (0.92)	3.23 (0.94)	3.17 (0.95)
Comp6	4.20 (0.55)	4.19 (0.54)	4.17 (0.56)

Supporting figures

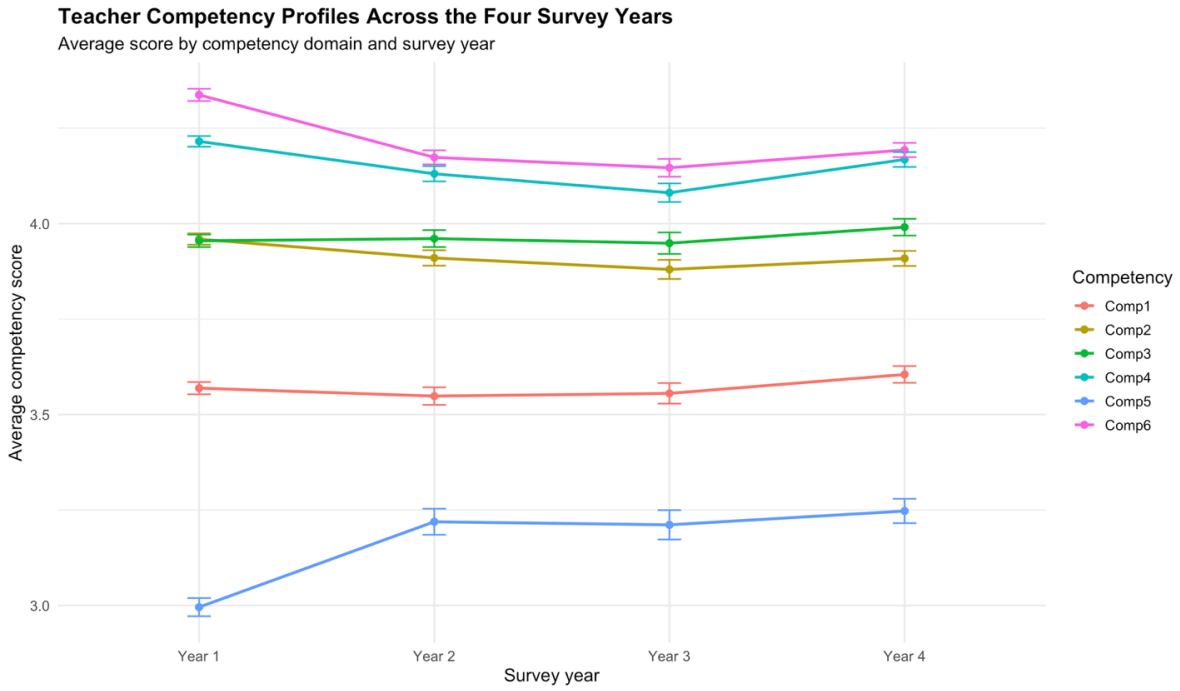
Appendix Figure B1. Year 4 teacher competency composite scores by grade level taught across the six domains.



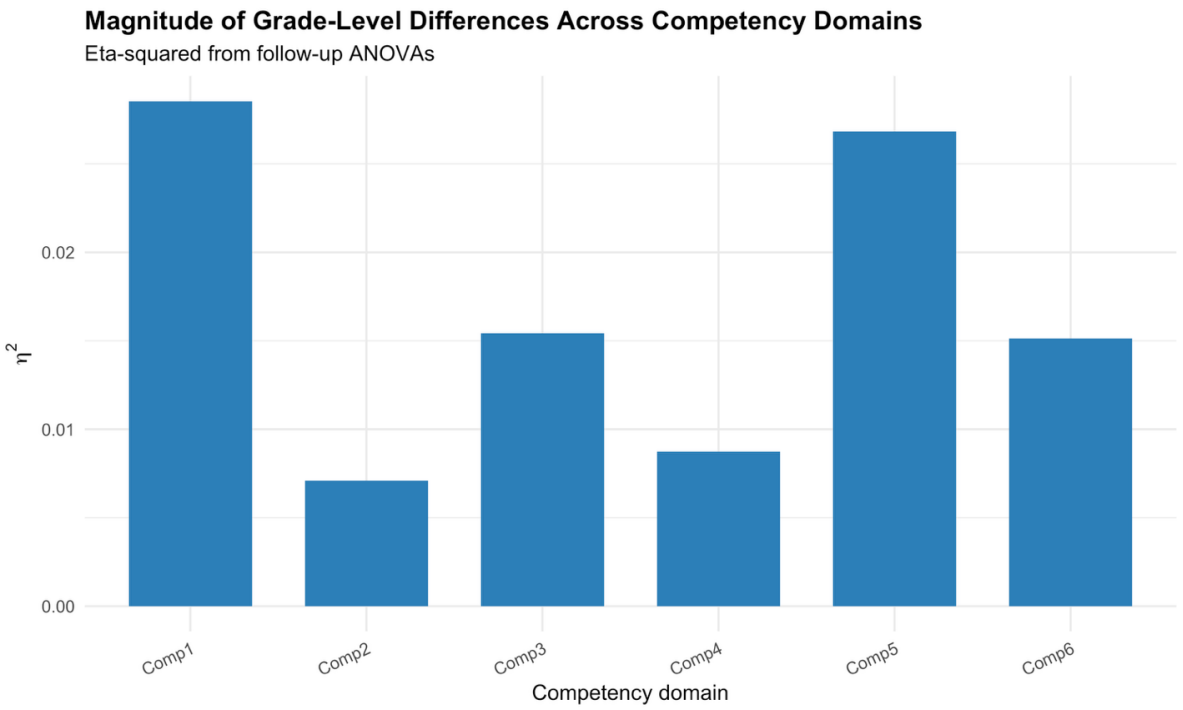
Appendix Figure B2. Year 4 teacher competency composite scores by teaching experience across the six domains.



Appendix Figure B3. Mean teacher competency profiles across the four survey years, showing a stable cross-year pattern with only modest movement in selected domains.



Appendix Figure B4. Eta-squared effect-size summary for grade-level differences across domains, illustrating that all detected contextual effects are small in magnitude.



Interpretation boundaries

- Interpretation depends on the correctness of the Year 4 grouping-variable coding and category labels.
- Some group categories may be small, so the detected differences should be read as targeted contextual signals rather than large profile separation.

Appendix C. Structural Alignment Between Implementation and Need

Domain-specific SEM evidence on whether stronger implementation aligns with lower support need

Overview

The structural analysis addresses a conceptually direct question: within each Teacher Quality Standard domain, is stronger reported implementation associated with lower reported professional learning need in that same domain? A structural equation modeling framework is used so that implementation and need are treated as latent constructs measured by multiple indicators rather than as simple observed averages.

Across the domains that could be estimated successfully, the structural path from implementation to need is consistently negative. Inclusive Environments shows the strongest inverse relationship, followed by Engaging in Career-Long Learning and Fostering Effective Relationships. The direction remains stable across sensitivity analyses, which strengthens the interpretation that stronger practice is aligned with lower perceived support need.

Measurement quality is strongest in Domains 2 and 4, while Domain 1 remains interpretable but somewhat weaker. Not all domains could be estimated under the current item mapping and data structure, which is itself an important analytical boundary that should remain visible in policy interpretation.

Analytical approach

- Domain-specific SEMs were fitted using Year 4 teacher data and two latent factors per estimable domain: implementation and professional learning need.
- The primary specification used ordered-data SEM; centered robust MLR and overlap-removed models were fitted as sensitivity checks.
- Model fit indices, factor loading ranges, and standardized structural paths were reviewed jointly before interpretation.

Supporting tables

Appendix Table C1. Standardized structural path estimates for the relationship between competency implementation and professional learning need.

Domain	Primary ordered SEM (WLSMV)	Sensitivity MLR centered	Sensitivity overlap removed
Domain 1: Fostering Effective Relationships	-0.17	-0.08	-0.17
Domain 2: Engaging in Career-Long Learning	-0.20	-0.17	-0.20
Domain 4: Inclusive Environments	-0.25	-0.22	-0.25

Appendix Table C2. Model fit indices for domain-specific SEMs under the primary and sensitivity models.

Domain	Model	CFI	TLI	RMSEA	SRMR	Overall interpretation
Domain 1: Fostering Effective Relationships	Primary WLSMV	0.984	0.979	0.090	0.072	Acceptable but weaker fit
Domain 2: Engaging in Career-Long Learning	Primary WLSMV	0.998	0.998	0.031	0.035	Excellent fit
Domain 4: Inclusive Environments	Primary WLSMV	0.995	0.991	0.063	0.041	Good fit

Appendix Table C3. Standardized factor loadings for competency implementation and professional learning need indicators.

Domain	Competency factor indicators	Competency loading range	Need factor indicators	Need loading range
Domain 1: Fostering Effective Relationships	Q5, Q4, Q81, Q82, Q83	0.61–0.78	Q44, Q105, Q106, Q107, Q108	0.63–0.94
Domain 2: Engaging in Career-Long Learning	Q84, Q85, Q86, Q87, Q88	0.57–0.72	Q109, Q110, Q113, Q114, Q116	0.75–0.93
Domain 4: Inclusive Environments	Q91, Q92, Q93	0.71–0.82	Q117, Q118, Q121, Q119	0.76–0.89

Appendix Table C4. Overlap between implementation and professional learning need indicators in the Year 4 appendix mapping.

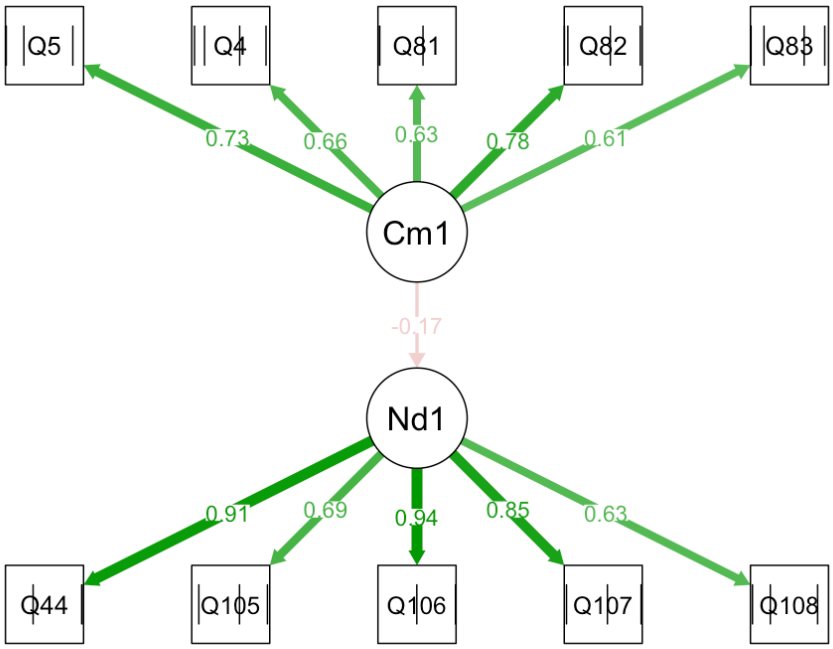
Domain	Competency block	Need block	Overlapping item(s)	Number of overlaps
Domain 1	Comp1	Need1	None	0
Domain 2	Comp2	Need2	None	0
Domain 3	Comp3	Need3	None detected in fitted structure	0
Domain 4	Comp4	Need4	Q95	1
Domain 5	Comp5	Need5	Q98	1
Domain 6	Comp6	Need6	None	0

Appendix Table C5. Summary interpretation of domain-specific SEM findings.

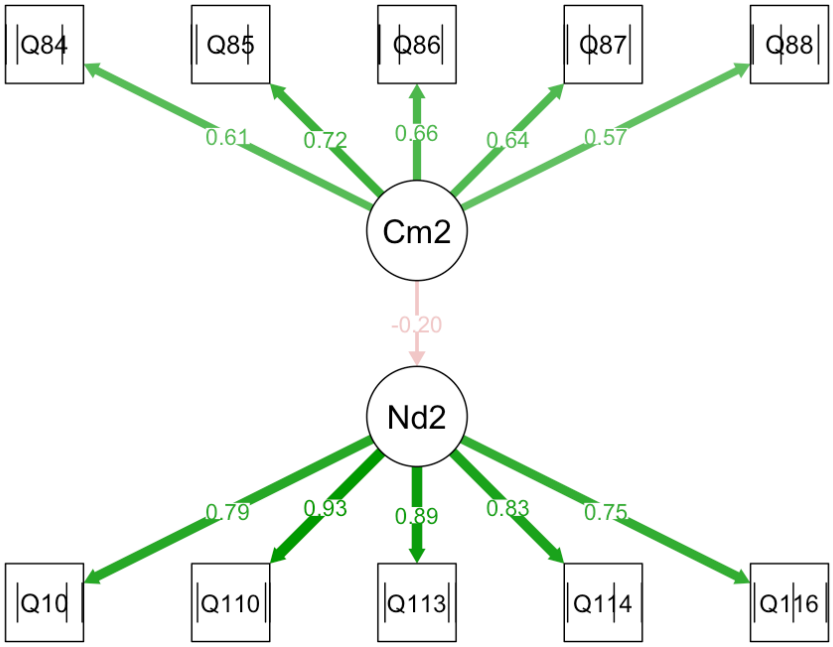
Domain	Structural direction	Approximate magnitude	Fit quality	Robustness across specifications	Overall interpretation
Domain 1: Fostering Effective Relationships	Negative	Weak-to-modest (-0.17 primary)	Acceptable but weaker	Direction stable, magnitude sensitive under centered MLR	Interpretable inverse relationship, but more caution warranted
Domain 2: Engaging in Career-Long Learning	Negative	Moderate (-0.20 primary)	Excellent	Highly stable across specifications	Strong and robust inverse relationship
Domain 4: Inclusive Environments	Negative	Moderate-to-strong (-0.25 primary)	Good	Highly stable, including overlap removal	Strongest and most robust inverse relationship

Supporting figures

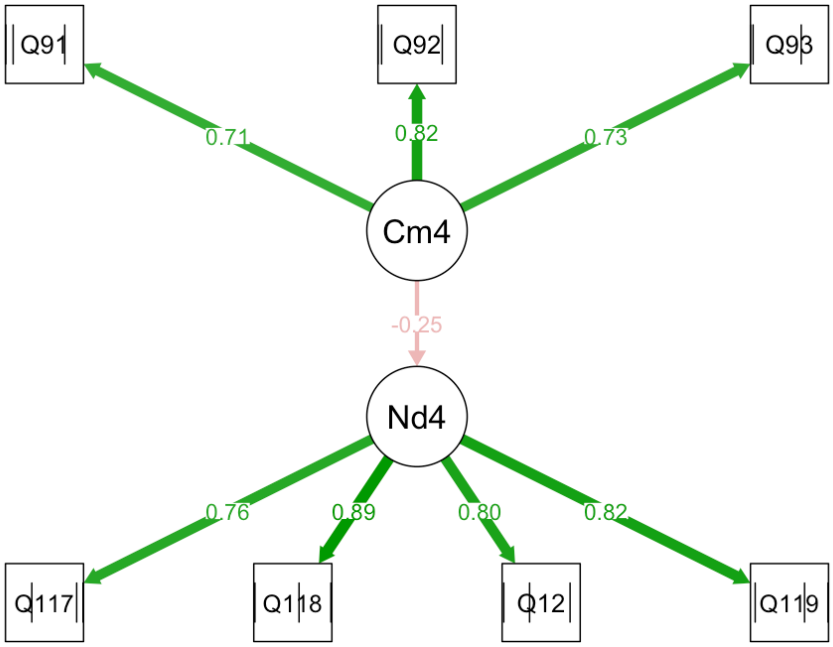
Appendix Figure C1. Structural equation model for Fostering Effective Relationships, showing the measurement structure for implementation and need indicators in Domain 1.



Appendix Figure C2. Structural equation model for Engaging in Career-Long Learning, showing the measurement structure for implementation and need indicators in Domain 2.



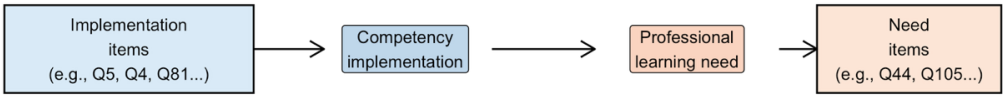
Appendix Figure C3. Structural equation model for Inclusive Environments, showing the measurement structure for implementation and need indicators in Domain 4.



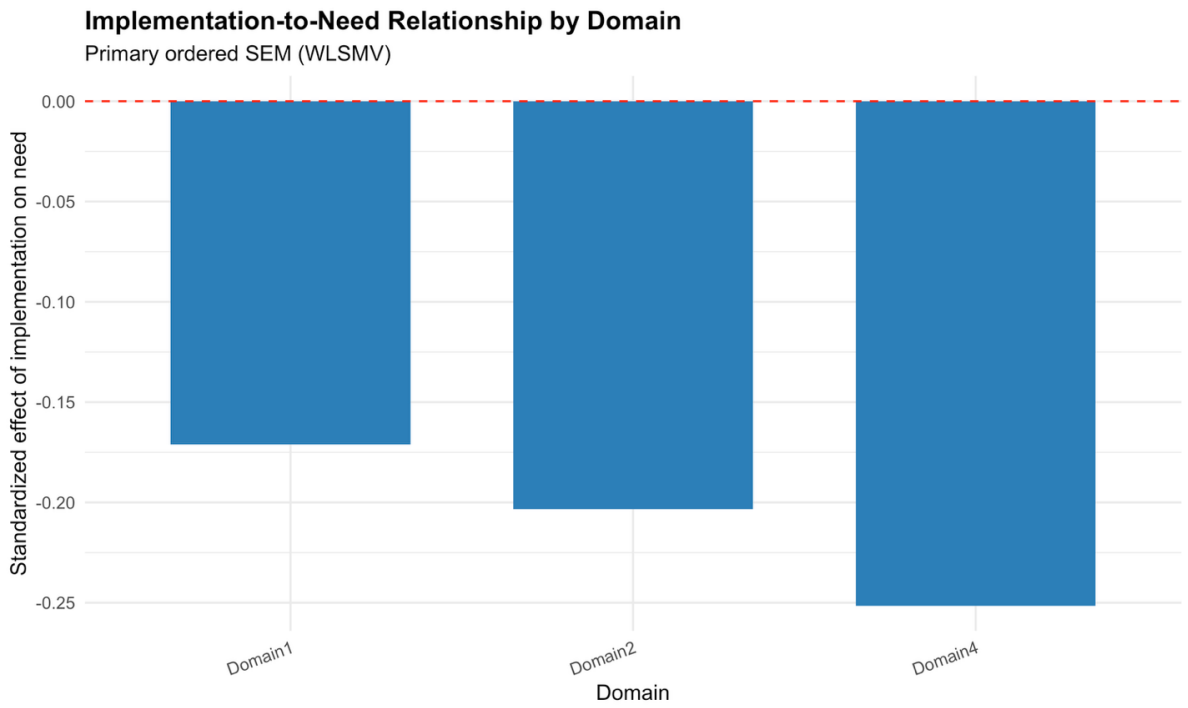
Appendix Figure C4. Conceptual schematic of the domain-specific SEM framework linking latent implementation to latent professional learning need.

Conceptual Structure of the Domain-Specific SEM

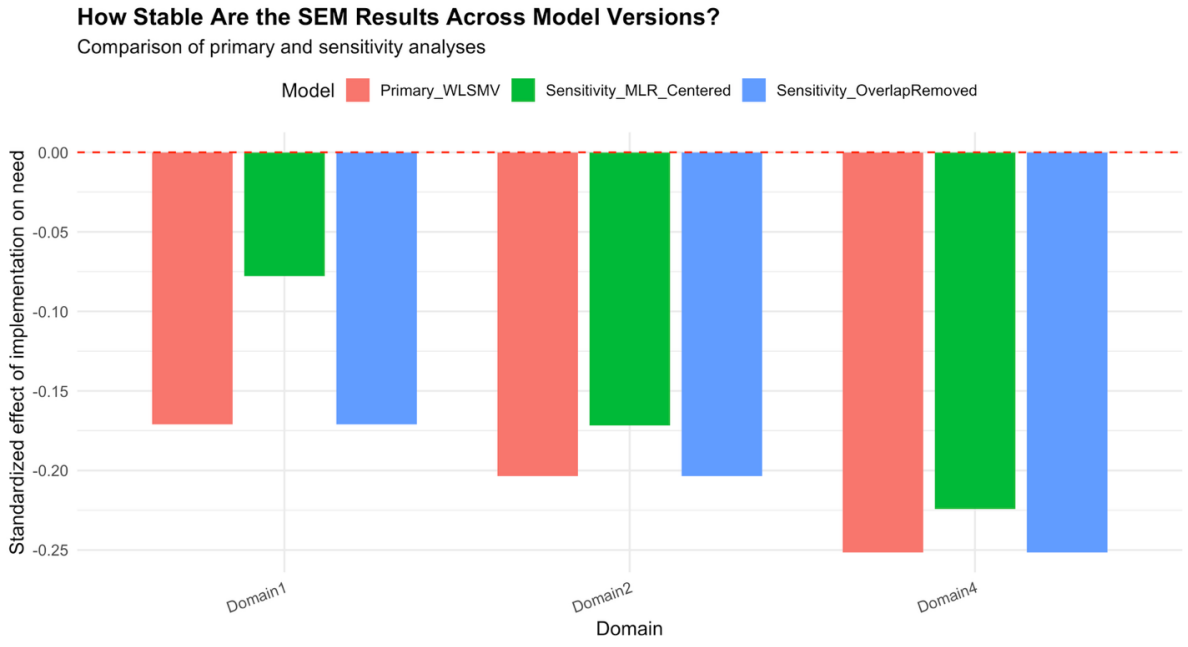
Each model links competency implementation to professional learning need



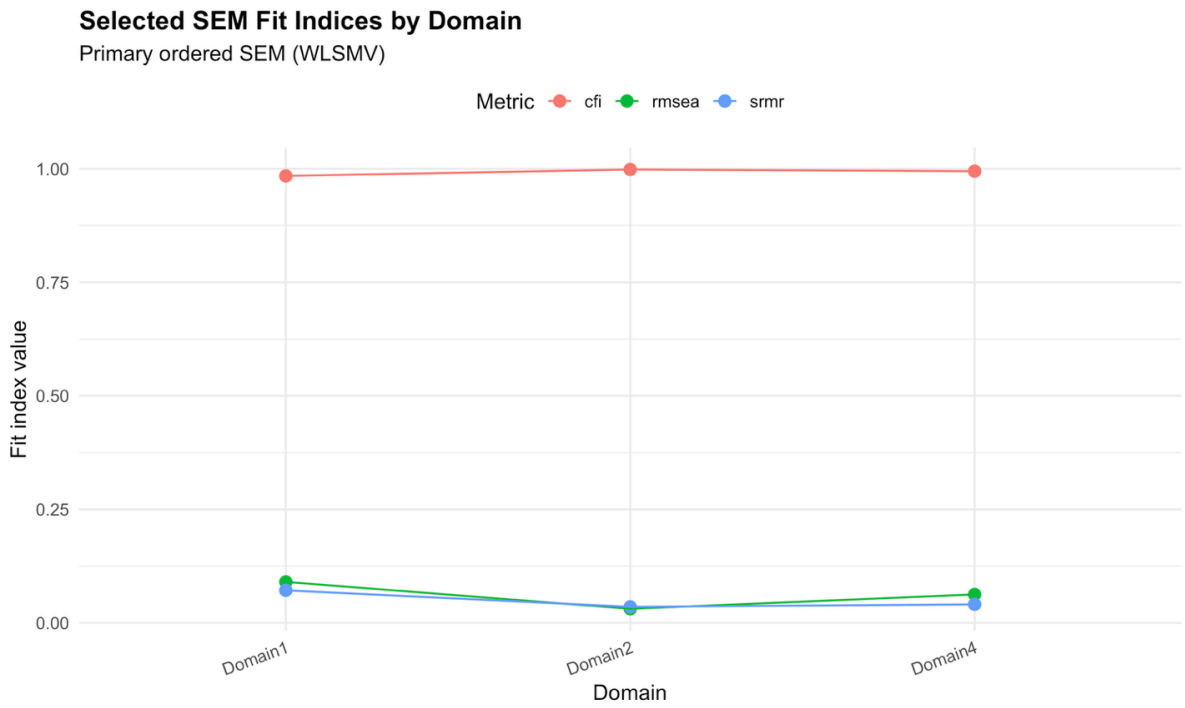
Appendix Figure C5. Primary ordered-SEM standardized paths from implementation to need by estimable domain.



Appendix Figure C6. Comparison of standardized implementation-to-need paths across SEM specifications, showing stable negative direction across models.



Appendix Figure C7. Selected SEM fit indices for the primary ordered models by domain.



Interpretation boundaries

- Only Domains 1, 2, and 4 were estimable under the current mapping and data constraints.
- Repeated question numbers across some implementation and need blocks should be confirmed before the SEM results are treated as fully final.

Appendix D. Longitudinal Context and Year 4 Positioning

How the Year 4 profile fits within the broader four-year implementation pattern

Overview

The longitudinal-context analysis places the Year 4 findings within the broader project trajectory. The available appendix documentation supports a four-year comparison of implementation, but not a defensible four-year trend analysis for professional learning need, because explicit need mappings are only available for Year 4.

Across Years 1-4, the implementation profile is highly stable. Legal Frameworks and Policies and Inclusive Environments remain among the strongest domains in every year, while FNMI Knowledge remains the weakest. Some domains show modest year-related movement, but the overall six-domain pattern is remarkably consistent over time.

Profile correlations between the Year 4 domain pattern and earlier years are extremely high, showing that Year 4 belongs to a stable system trajectory rather than representing a new implementation landscape. The longitudinal evidence therefore strengthens the interpretation that the report is identifying enduring strengths and persistent needs rather than transient one-year effects.

Analytical approach

- Implementation domains were scored separately within each year using the documented year-specific mappings and the same minimum-completion rule used elsewhere in the report.
- Domain-specific ANOVAs tested whether mean implementation differed across years, with multiple-testing correction applied across domains.
- Year 4 consistency with earlier years was examined through domain rank ordering and correlations of the Year 4 mean profile with earlier yearly mean profiles.
- Year 4 professional learning need was summarized descriptively only, because multi-year need mappings were not available.

Supporting tables

Appendix Table D1. Longitudinal implementation trend summary by year and Teacher Quality Standard domain.

Domain	Year 1	Year 2	Year 3	Year 4
Comp1: Fostering Effective Relationships	3.57 (0.67)	3.55 (0.66)	3.56 (0.63)	3.61 (0.64)
Comp2: Engaging in Career-Long Learning	3.96 (0.62)	3.91 (0.58)	3.88 (0.58)	3.91 (0.57)
Comp3: Professional Body of Knowledge	3.96 (0.69)	3.96 (0.63)	3.94 (0.66)	3.99 (0.64)
Comp4: Inclusive Environments	4.21 (0.59)	4.13 (0.57)	4.08 (0.56)	4.17 (0.57)
Comp5: FNMI Knowledge	3.00 (1.00)	3.22 (0.97)	3.21 (0.89)	3.25 (0.93)
Comp6: Legal Frameworks & Policies	4.34 (0.67)	4.17 (0.53)	4.15 (0.54)	4.19 (0.55)

Appendix Table D2. Full implementation summary with sample size, median, and quartiles.

Year	Domain	n	Mean	SD	Median	Q1	Q3
Year 1	Comp1: Fostering Effective Relationships	1783	3.57	0.67	3.67	3.00	4.00

Year	Domain	n	Mean	SD	Median	Q1	Q3
Year 1	Comp2: Engaging in Career-Long Learning	1782	3.96	0.62	4.00	3.50	4.50
Year 1	Comp3: Professional Body of Knowledge	1780	3.96	0.69	4.00	3.50	4.50
Year 1	Comp4: Inclusive Environments	1780	4.21	0.59	4.00	4.00	4.50
Year 1	Comp5: FNMI Knowledge	1776	3.00	1.00	3.00	2.00	3.75
Year 1	Comp6: Legal Frameworks & Policies	1776	4.34	0.67	4.00	4.00	5.00
Year 2	Comp1: Fostering Effective Relationships	822	3.55	0.66	3.60	3.00	4.00
Year 2	Comp2: Engaging in Career-Long Learning	821	3.91	0.58	4.00	3.60	4.40
Year 2	Comp3: Professional Body of Knowledge	818	3.96	0.63	4.00	3.50	4.50
Year 2	Comp4: Inclusive Environments	816	4.13	0.57	4.00	3.92	4.67
Year 2	Comp5: FNMI Knowledge	819	3.22	0.97	3.00	2.67	4.00
Year 2	Comp6: Legal Frameworks & Policies	816	4.17	0.53	4.00	4.00	4.67
Year 3	Comp1: Fostering Effective Relationships	541	3.56	0.63	3.60	3.20	4.00
Year 3	Comp2: Engaging in Career-Long Learning	540	3.88	0.58	3.80	3.40	4.20
Year 3	Comp3: Professional Body of Knowledge	538	3.94	0.66	4.00	3.50	4.50
Year 3	Comp4: Inclusive Environments	537	4.08	0.56	4.00	3.67	4.33
Year 3	Comp5: FNMI Knowledge	539	3.21	0.89	3.00	2.67	4.00
Year 3	Comp6: Legal Frameworks &	537	4.15	0.54	4.00	4.00	4.67

Year	Domain	n	Mean	SD	Median	Q1	Q3
	Policies						
Year 4	Comp1: Fostering Effective Relationships	858	3.61	0.64	3.60	3.20	4.00
Year 4	Comp2: Engaging in Career-Long Learning	857	3.91	0.57	4.00	3.60	4.40
Year 4	Comp3: Professional Body of Knowledge	855	3.99	0.64	4.00	3.50	4.50
Year 4	Comp4: Inclusive Environments	851	4.17	0.57	4.00	4.00	4.67
Year 4	Comp5: FNMI Knowledge	854	3.25	0.93	3.00	3.00	4.00
Year 4	Comp6: Legal Frameworks & Policies	851	4.19	0.55	4.00	4.00	4.67

Appendix Table D3. Domain-specific ANOVA results for year differences in implementation scores.

Domain	F statistic	Raw p-value
Comp1: Fostering Effective Relationships	1.227	0.2983
Comp2: Engaging in Career-Long Learning	3.186	0.02284
Comp3: Professional Body of Knowledge	0.826	0.4792
Comp4: Inclusive Environments	9.042	5.785e-06
Comp5: FNMI Knowledge	19.198	2.343e-12
Comp6: Legal Frameworks & Policies	24.830	6.556e-16

Appendix Table D4. Multiple-testing-adjusted year effects across implementation domains.

Domain	F statistic	Raw p-value	BH-adjusted p	Holm-adjusted p
Comp1: Fostering Effective Relationships	1.227	0.2983	0.3580	0.5967
Comp2: Engaging in Career-Long Learning	3.186	0.02284	0.03426	0.06852
Comp3: Professional Body of Knowledge	0.826	0.4792	0.4792	0.5967
Comp4: Inclusive Environments	9.042	5.785e-06	1.157e-05	2.314e-05
Comp5: FNMI Knowledge	19.198	2.343e-12	7.030e-12	1.172e-11
Comp6: Legal Frameworks & Policies	24.830	6.556e-16	3.934e-15	3.934e-15

Appendix Table D5. Rank ordering of implementation domains within each survey year.

Rank within year	Year 1	Year 2	Year 3	Year 4
1	Comp6: Legal Frameworks & Policies	Comp6: Legal Frameworks & Policies	Comp6: Legal Frameworks & Policies	Comp6: Legal Frameworks & Policies
2	Comp4: Inclusive Environments	Comp4: Inclusive Environments	Comp4: Inclusive Environments	Comp4: Inclusive Environments
3	Comp2: Engaging in Career-Long Learning	Comp3: Professional Body of Knowledge	Comp3: Professional Body of Knowledge	Comp3: Professional Body of Knowledge
4	Comp3: Professional Body of Knowledge	Comp2: Engaging in Career-Long Learning	Comp2: Engaging in Career-Long Learning	Comp2: Engaging in Career-Long Learning
5	Comp1: Fostering Effective Relationships	Comp1: Fostering Effective Relationships	Comp1: Fostering Effective Relationships	Comp1: Fostering Effective Relationships
6	Comp5: FNMI Knowledge	Comp5: FNMI Knowledge	Comp5: FNMI Knowledge	Comp5: FNMI Knowledge

Appendix Table D6. Correlation of the Year 4 domain profile with earlier yearly implementation profiles.

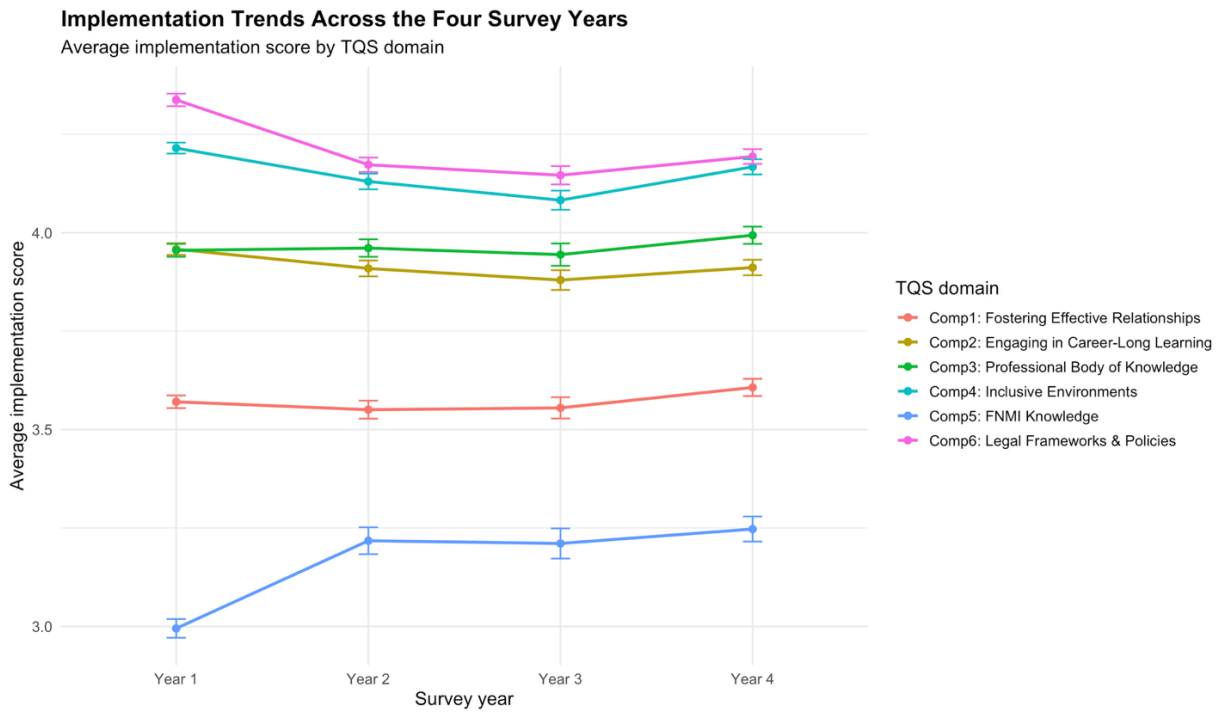
Comparison	Correlation
Year4 vs Year1	0.9936
Year4 vs Year2	0.9989
Year4 vs Year3	0.9990

Appendix Table D7. Year 4 professional learning need summary by domain.

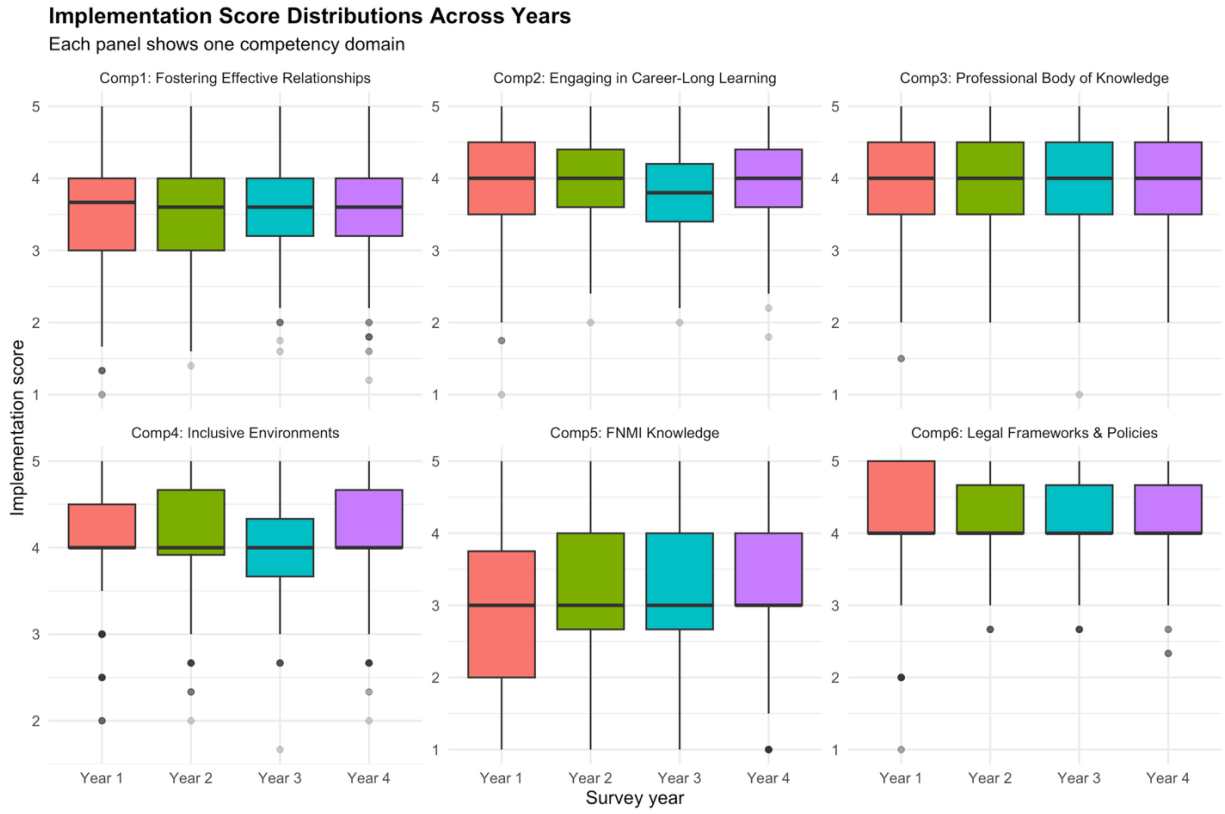
Need domain	n	Mean	SD	Median
Need1	736	2.25	0.72	2.20
Need2	735	2.21	0.74	2.20
Need3	735	2.33	0.81	2.33
Need4	735	2.34	0.74	2.25
Need5	735	2.67	0.77	3.00
Need6	735	2.04	0.87	2.00

Supporting figures

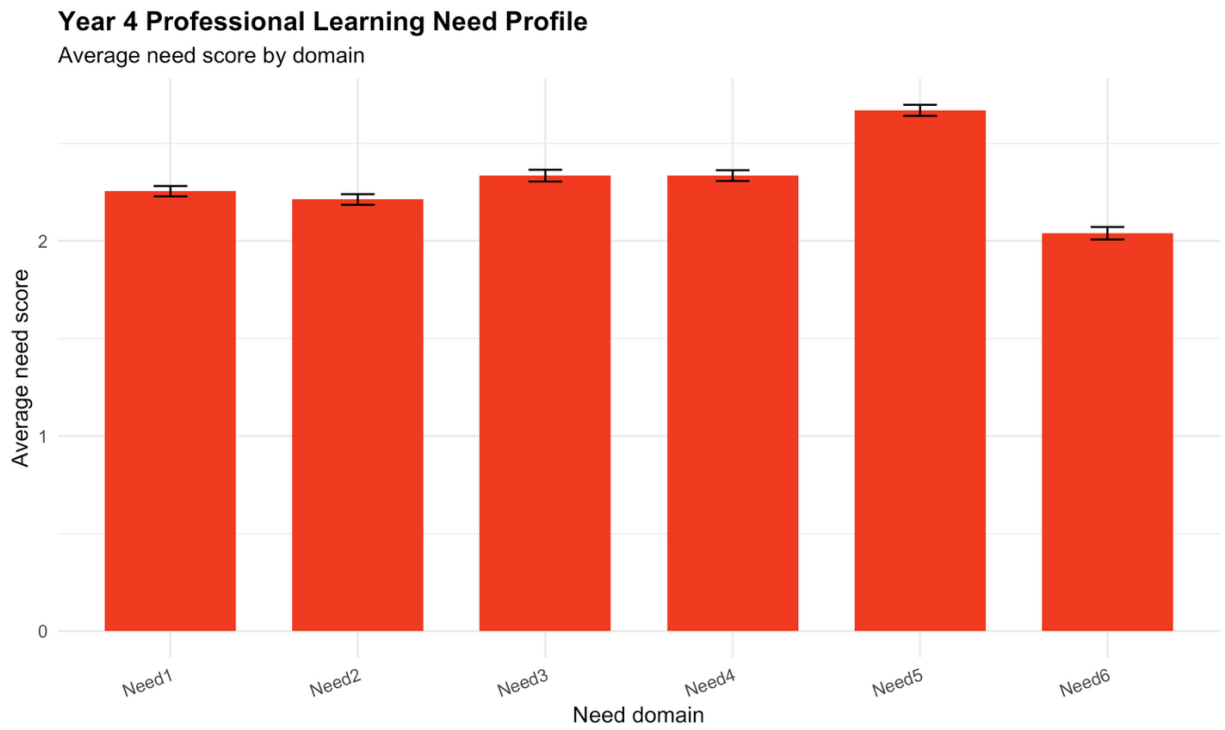
Appendix Figure D1. Implementation trends across the four survey years by domain.



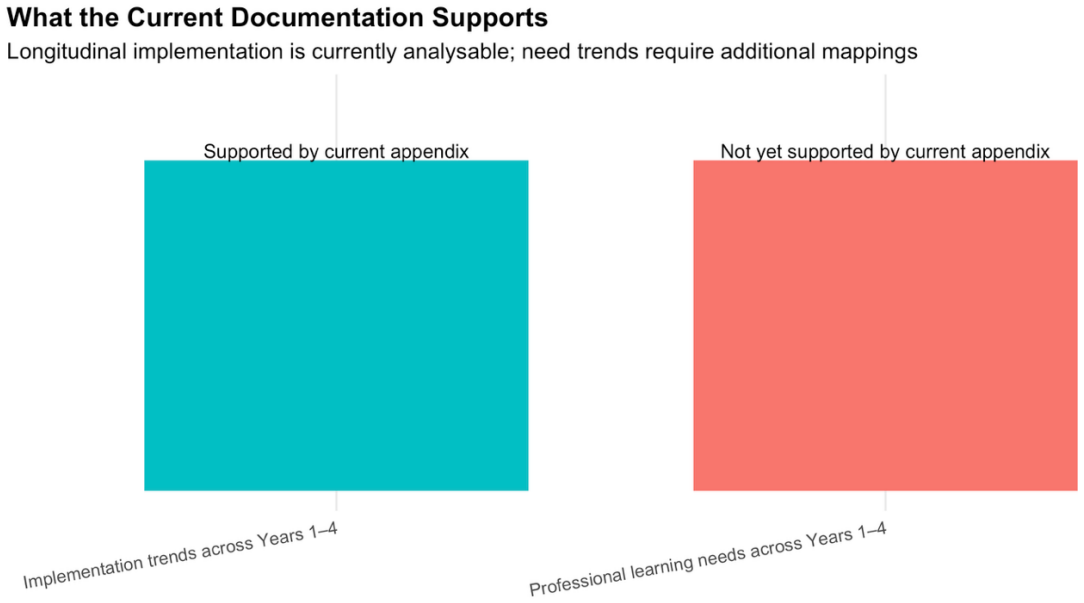
Appendix Figure D2. Distribution of implementation scores across years within each domain.



Appendix Figure D3. Year 4 professional learning need profile by domain.



Appendix Figure D4. Current documentation support for longitudinal analyses, showing what can be interpreted confidently from the available mappings.



Interpretation boundaries

- These results are best interpreted as repeated cross-sectional trend evidence rather than within-person longitudinal change.
- Cross-year comparisons remain subject to partial mapping differences, especially between Year 1 and later waves.